

**Mansoura University
Faculty of Computers
and Information
Department of
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Agile Methods

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Lecture 10

Documentation of Data and Process Models

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Documentation Of Data and Process Models

Data Dictionary

- Using CASE Tools for Documentation

Process Description Tools

- Modular Design
- Structured English
- Decision Table
- Decision Trees

Logical versus Physical Models

- Sequence of Models
- Four-Model Approach



Data Dictionary

Data Dictionary outlining a Database on Driver Details in NSW

Field Name	Data Type	Data Format	Field Size	Description	Example
License ID	Integer	NNNNNN	6	Unique number ID for all drivers	12345
Surname	Text		20	Surname for Driver	Jones
First Name	Text		20	First Name for Driver	Arnold
Address	Text		50	First Name for Driver	11 Rocky st Como 2233
Phone No.	Text		10	License holders contact number	0400111222
D.O.B	Date / Time	DD/MM/YYYY	10	Drivers Date of Birth	08/05/1956

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Data Dictionary

A set of DFDs produces a **logical model** of the system, but the **details** within those DFDs are documented **separately** in a data dictionary, which is the **second component** of structured analysis.

A **data dictionary**, or **data repository**, is a central storehouse of information about the system's data.

Data Dictionary

Data Dictionary outlining a Database on Driver Details in NSW

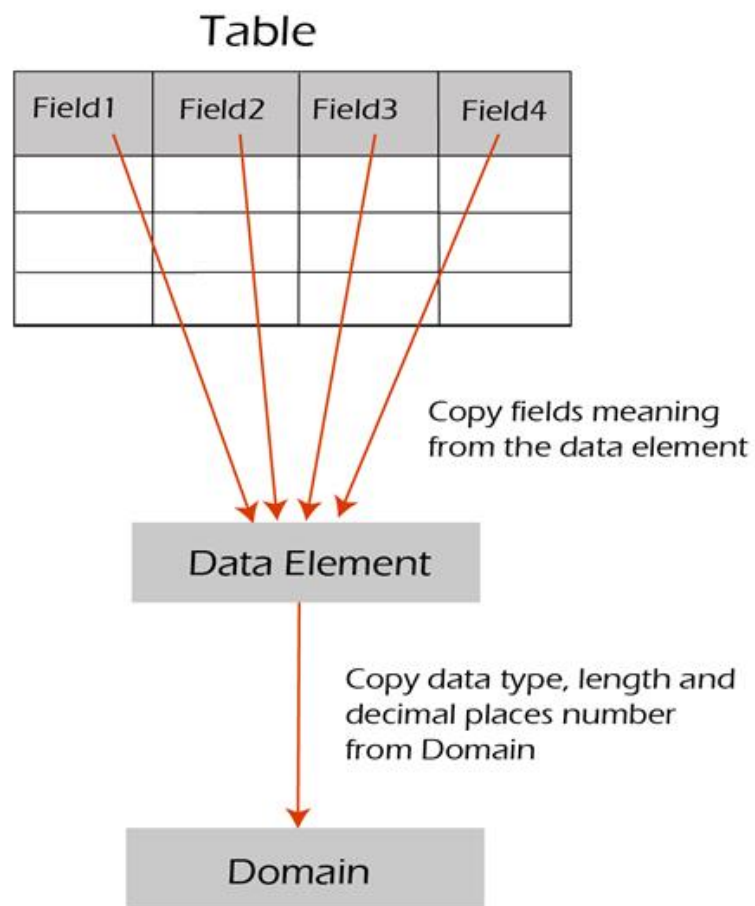
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Data Dictionary

An analyst uses the data dictionary to collect, document, and organize specific facts about the system, including the contents of data flows, data stores, entities, and processes.

The data dictionary also defines and describes all data elements and meaningful combinations of data elements.



Data Element

A **data element**, also called a data item or field, is the smallest piece of data that has meaning within an information system.

The **Domain** is the place that provide the data elements.

Examples of data elements are student grade, salary, Social Security number, account balance, and company name. Data elements are combined into records, also **called data structures**.

USING CASE TOOLS FOR DOCUMENTATION

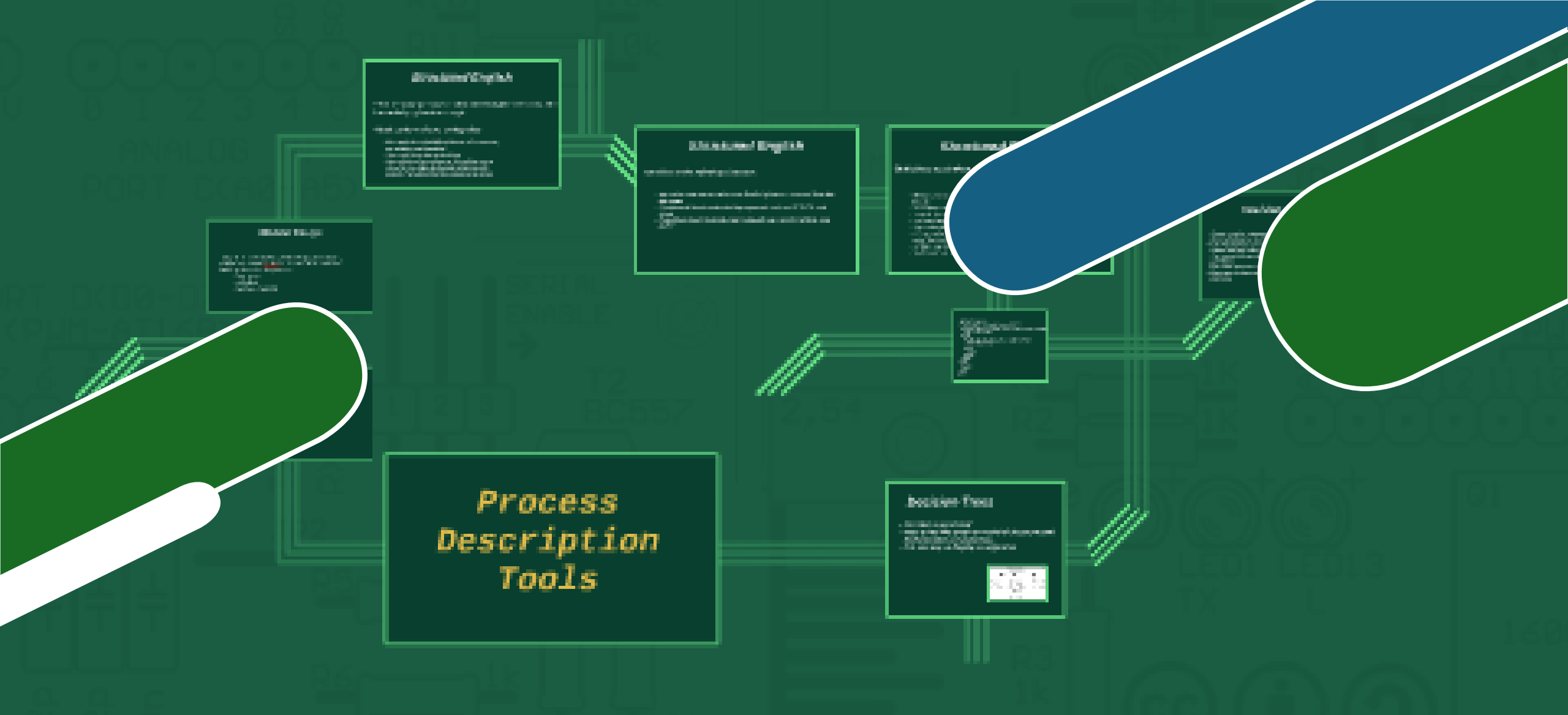
- The more **complex** the system, the more **difficult** it is to maintain full and accurate documentation.
- Fortunately, modern CASE tools **simplify** the task by flowing documentation **automatically** from the **modeling diagrams** into the **central repository**, along with information entered by the user.



USING CASE TOOLS FOR DOCUMENTATION

- A CASE repository ensures **data consistency**, which is especially important where multiple systems require the same data.
- In a large company, for example, the sales, accounting, and shipping systems all might use a data element called **CUSTOMER NUMBER (SSN)**.
- Once the CUSTOMER NUMBER element has been defined in the repository, other processes can access it, data flows, and data stores.

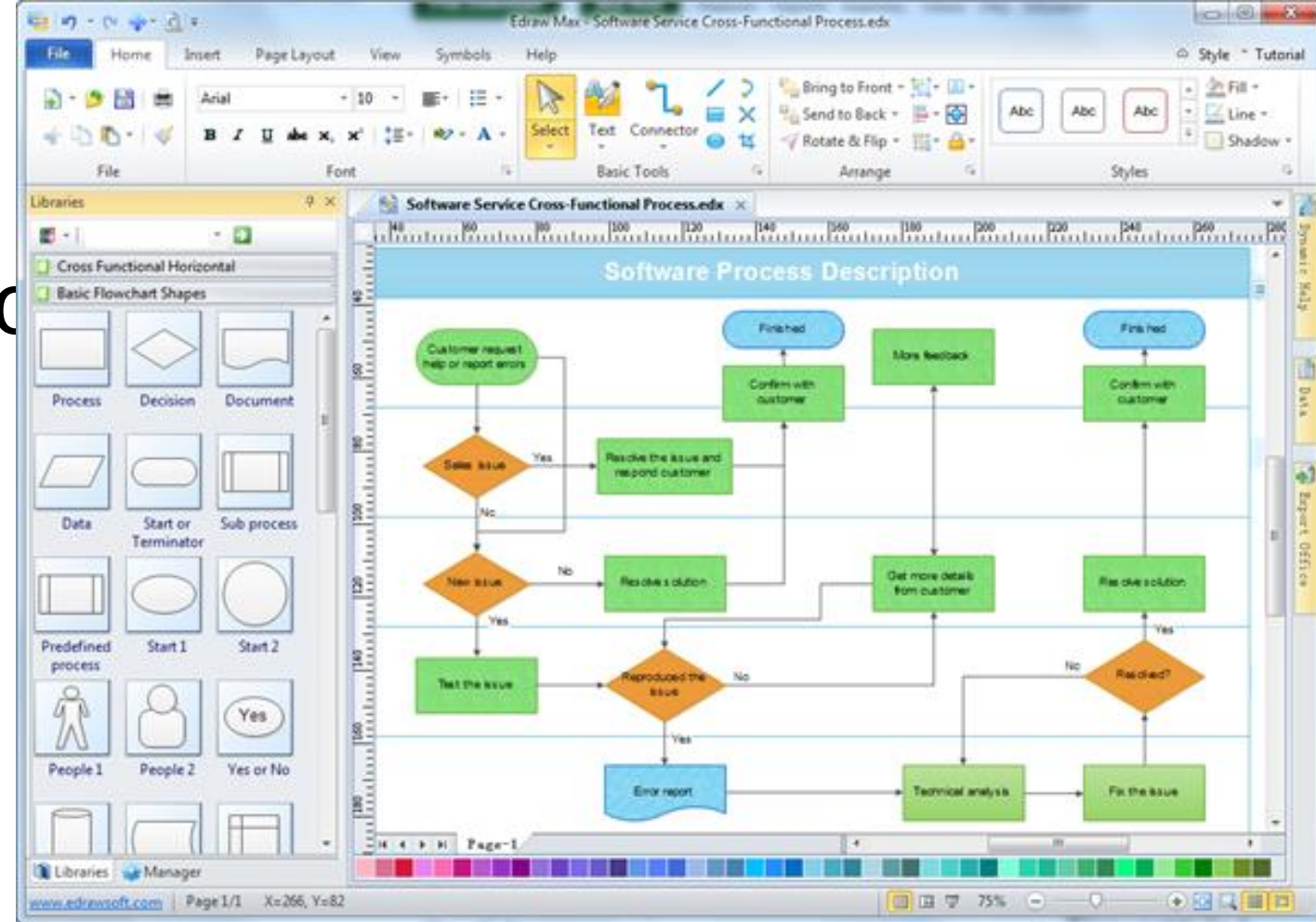
The screenshot shows a software interface for defining a data element. It has a tabbed menu at the top with 'Description' selected. The 'Label' field contains 'SOCIAL SECURITY NUMBER' and a '1 of 4' indicator is on the right. The 'Entry Type' is set to 'Data Element'. The 'Description' field contains 'Social Security Number'. The 'Alias' field contains 'SSN'. The 'Values & Meanings' section is expanded, showing 'Type and length: 9N', 'Default value: None', and 'Acceptable values: Any nine digit number'. The 'Notes' section contains 'Source: Application form', 'Security: Payroll department', and 'Responsible user: Payroll department'. The 'Long Name' field is empty. At the bottom, there is a row of buttons: 'SQL', 'Delete', 'Next', 'Save' (highlighted in blue), 'Search', 'Jump', 'File', 'History', and '?'. Below these are more buttons: 'Dialect...', 'Clear', 'Prior', 'Exit', 'Expand', 'Back', 'Copy', and 'Search Criteria'. A footer note states: 'A repository object label can be up to 128 characters long, and the first character must be a letter.'



A **process description** documents the details of a **functional primitive** and represents a specific set of processing steps and business logic.

Process Description Tools

- Using a set of process description tools, a model is created that is accurate, complete, and concise.
- Typical process description tools include **structured English, decision tables, and decision trees**.
- When a functional primitive is analyzed, the processing steps are broken down into smaller units in a process called **modular design**.



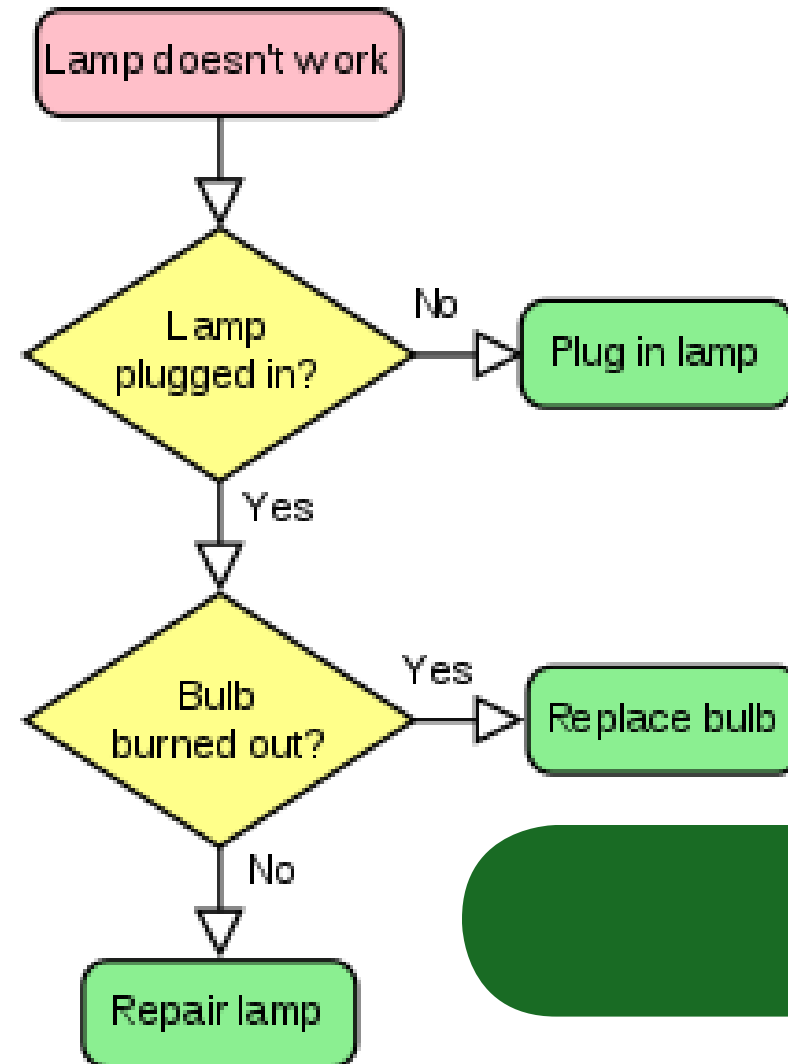
MODULAR DESIGN

- **Modular design** is based on combinations of three logical structures, sometimes called **control structures**, which serve as building blocks for the process.
- **Each logical structure** must have a single entry and exit point.
- The three structures are **called sequence, selection, and iteration.**



MODULAR DESIGN

- You really know the flow charts
- A rectangle represents a step or process.
- A diamond shape represents a condition or decision
- Logic follows the lines in the direction indicated by the arrows.



SEQUENCE

```
Console.WriteLine($"=====Sequence Implementation=====");  
#region Sequence Implementation  
VerifyProductCode();  
VerifyPrice();  
VerifyStockLevel();  
#endregion
```



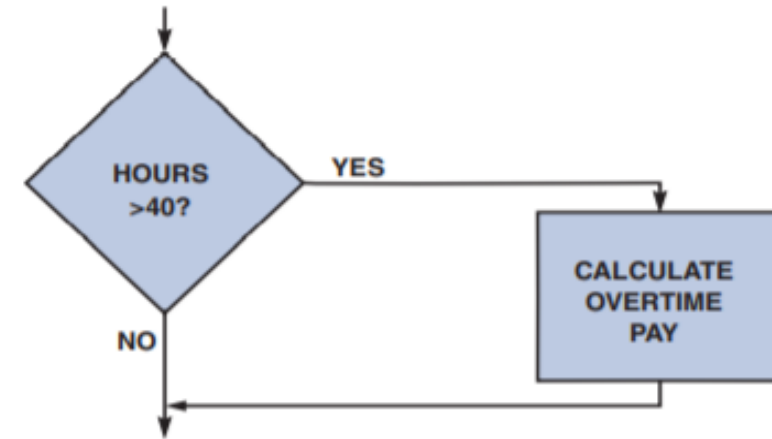
- The completion of steps in sequential order, one after another.
- One or more of the steps might represent a subprocess that contains additional logical structures.

```
> dotnet run
```

```
=====Sequence Implementation=====  
Product Code Verified  
Price Verified  
Stock level Verified
```

SELECTION.

```
Console.WriteLine($"=====Selection Implementation=====");  
#region Selection Implementation  
var Hours = 41;  
if(Hours >40){  
    CalculateOvertimePay();  
}  
#endregion
```

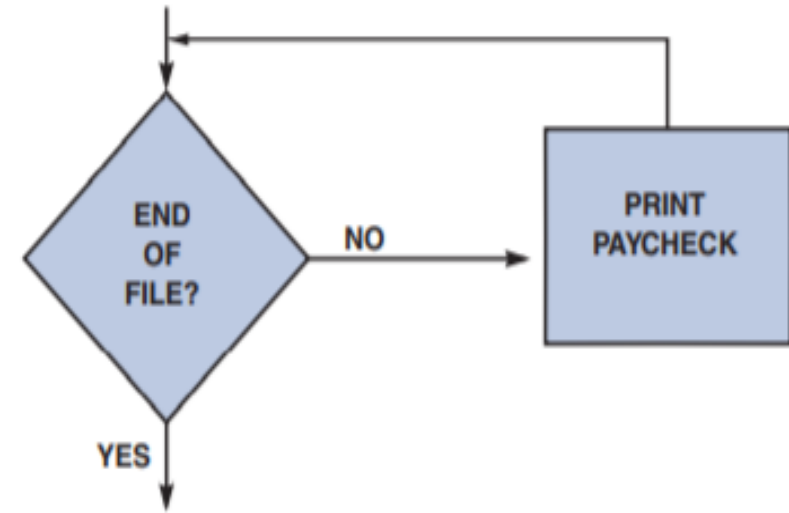


```
=====Selection Implementation=====  
Overtime pay = 20
```

- The completion of one of two or more process steps based on the results of a test or condition.
- The system tests the input, and if the hours are greater than 40, it performs the CALCULATE OVERTIME PAY process.

Iteration.

```
Console.WriteLine($"=====Iteration Implementation=====");  
#region Iteration Implementation  
var line = 0;  
var end = 10;  
for ( ; !EndOfFile(ref line, end); )  
{  
    Console.WriteLine("File no end");  
}  
CheckPay();  
#endregion
```



```
=====Iteration Implementation=====  
File no end * File no end * File no end * File no end * File no end * File no end * File no end * File no end *  
no end *  
Pay Checked
```

- The completion of a process step that is repeated until a specific condition changes.
- An example of iteration is a process that continues to print **File No End** until it reaches the end of the payroll file.
- Iteration also is called **looping**
- Sequence, selection, and iteration structures can be combined in various ways to describe processing logic.

Structured English

- Structured English is a subset of standard English that describes **logical processes** clearly and accurately. When using structured English, be mindful of the following guidelines:
 - Use only the three building blocks of sequence, selection, and iteration.
 - Use indentation for readability.
 - Use a limited vocabulary, including standard terms used in the data dictionary and specific words that describe the processing rules
- Structured English might look familiar to programming students because it resembles **pseudocode**

```
Follow route
Repeat
  IF is the light green?
    Repeat
      Keep moving to landmark
    Until it reached the landmark
    IF is there a treasure at the landmark
      Display 'treasure found'
      Collect Treasure
    Else
      Display 'no treasure'
      Move to another landmark
  Else
    IF is the light amber
      Wait (3 seconds)
    Else
      IF is the light red
        Repeat
          Stop
        Until the light changes
  Until all the treasure have been found
End of route
```

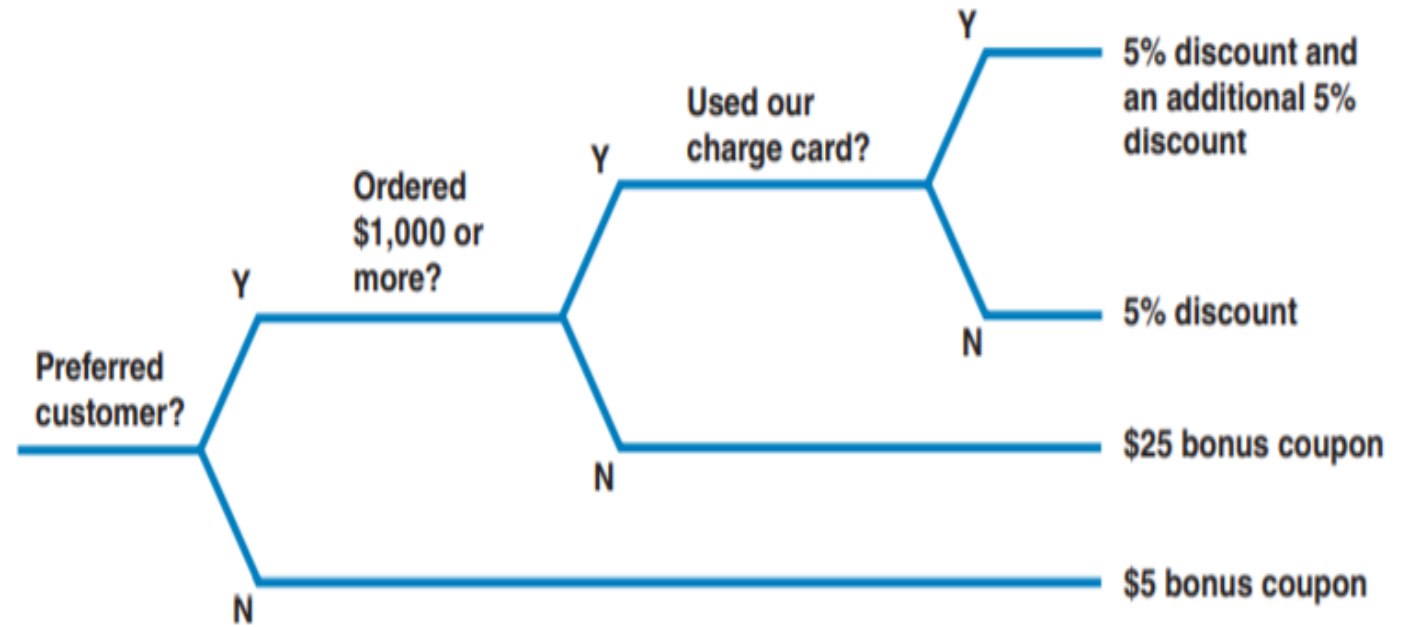
DECISION TABLE

- A decision table is a logical structure that shows every combination of conditions and outcomes.
- Analysts often use decision tables to describe a process and ensure that they have considered all possible situations.

	1	2	3	4	5	6	7	8
Preferred customer	Y	Y	Y	Y	N	N	N	N
Ordered \$1,000 or more	Y	Y	N	N	Y	Y	N	N
Used our charge card	Y	N	Y	N	Y	N	Y	N
5% discount	X	X						
Additional 5% discount	X							
\$25 bonus coupon			X	X				
\$5 bonus coupon					X	X	X	X

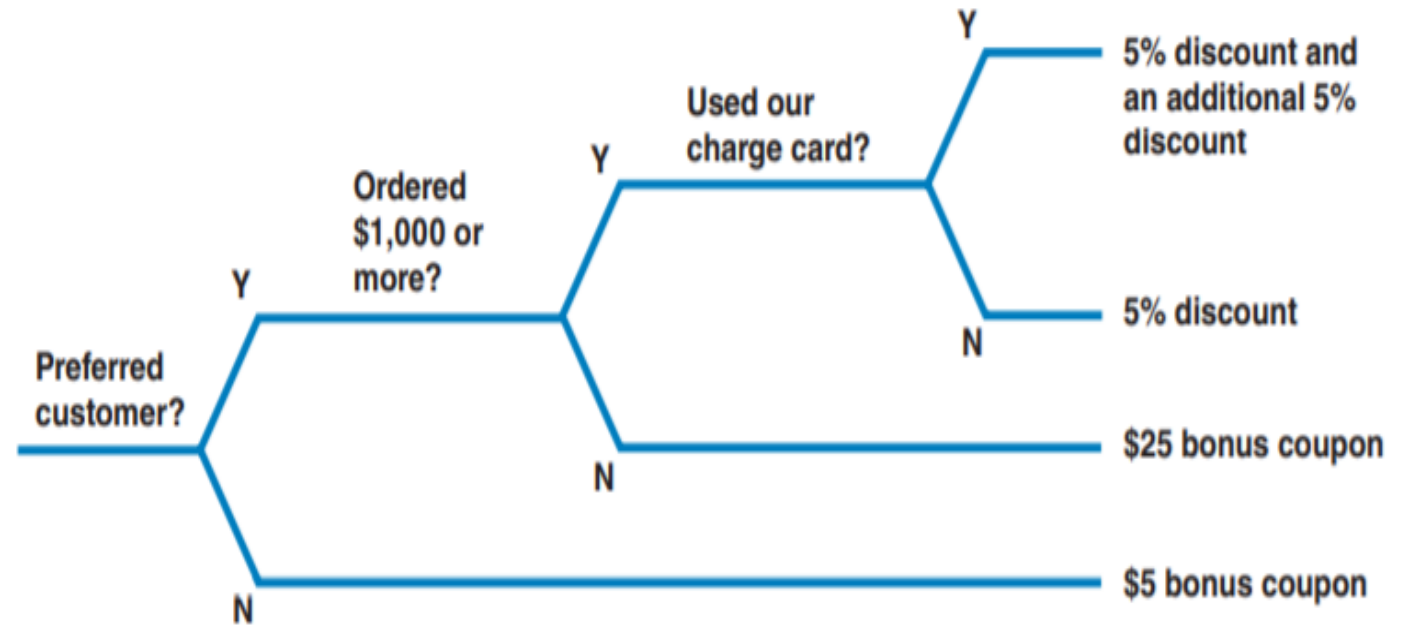
DECISION TREES

- A **decision tree** is a graphical representation of the conditions, actions, and rules found in a **decision table**.
- Decision trees show the logic structure in a **horizontal** form that resembles a tree with the roots at the left and the branches to the right.



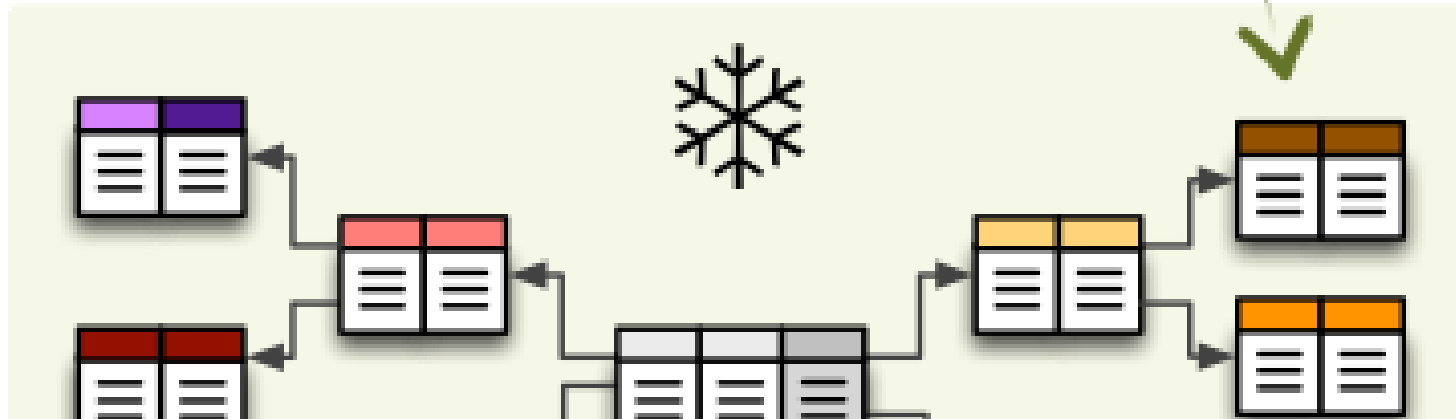
DECISION TREES

- Like flowcharts, decision trees are useful ways to present the system to management.
- Decision trees and decision tables provide the same results, but in different forms.
- In many situations, a graphic is the most effective means of communication.



- open procedure
- unknown
- competitive dialogue
- restricted procedure
- negotiated procedure without publishing

physical



Logical Vs Physical Model

Logical Vs Physical Model

- While structured analysis tools are used to develop a **logical model** for a new information system, such tools can also be used to develop **physical models** of an information system.
- A **physical model** shows how the system's requirements are implemented.

Feature	Conceptual	Logical	Physical
Entity Names	✓	✓	
Entity Relationships	✓	✓	
Attributes		✓	
Primary Keys		✓	✓
Foreign Keys		✓	✓
Table Names			✓
Column Names			✓
Column Data Types			✓

Logical Vs Physical Model

- During the systems design phase, a **physical model** of the new information system is created that follows from the logical model and involves operational tasks and techniques.

Feature	Conceptual	Logical	Physical
Entity Names	✓	✓	
Entity Relationships	✓	✓	
Attributes		✓	
Primary Keys		✓	✓
Foreign Keys		✓	✓
Table Names			✓
Column Names			✓
Column Data Types			✓

SEQUENCE OF MODELS

- To understand the relationship between logical and physical models, think back to the beginning of the systems analysis phase.
- To understand how the current tasks were carried out the existing system, the physical operations of the existing system were studied before the logical model.



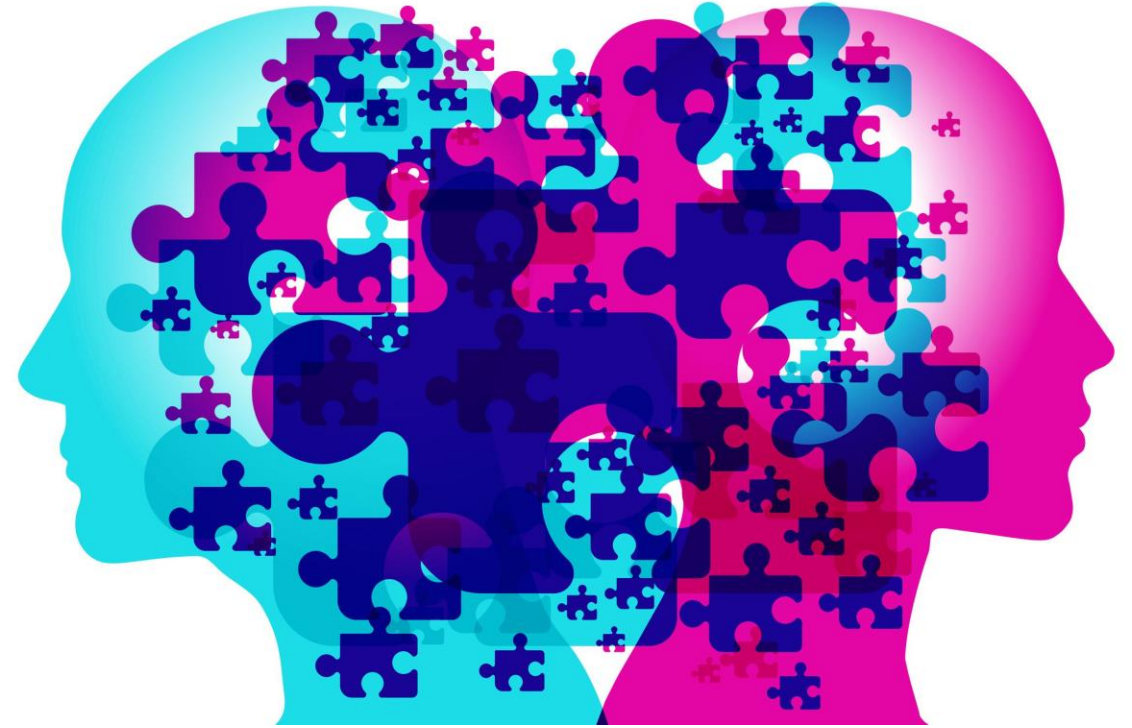
SEQUENCE OF MODELS

- Many systems analysts create a **physical model** of the current system and then develop a **logical model** of the current system before **tackling** a **logical model** of the new system.
- Performing that extra step allows them to **understand** the current system better.



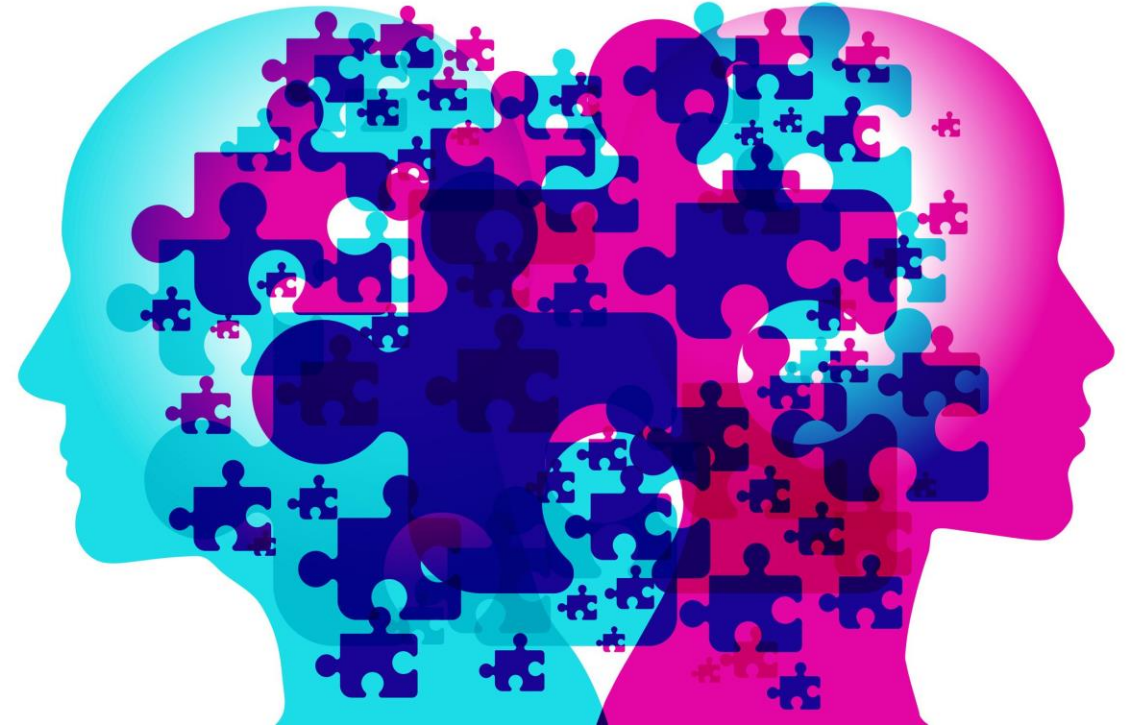
FOUR-MODEL APPROACH

- Many analysts follow a four-model approach,
- which means that they develop a physical model of the current system.
- Then a logical model of the current system
- Then a logical model of the new system. Finally a physical model of the new system.



FOUR-MODEL APPROACH

- Major benefit of the four-model approach is that it provides a clear picture of current system functions before any modifications or improvements are made.
- The only disadvantage of the four-model approach is the added time and cost needed to develop a logical and physical model of the current system.





THANK YOU!

